

Smart Metro Ticket Booking System

PROJECT PLANNING PHASE

Date	
Team ID	
Project Name	Smart Metro Ticket Booking System
Maximum Marks	4 Marks

2. Project Planning

2.1 Objective

This phase translates the finalized MVP scope into an actionable project plan. The work is organized into sprints, each functional requirement is expressed as a user story with an assigned priority and story point estimate, and a realistic delivery timeline is established for building the solution on the ServiceNow platform.

2.2 Sprint Planning

Sprint	Functional Requirement	User Story	Points	Priority
Sprint 1	Station & Booking Catalog	As a commuter, I want to select source, destination, date, and passenger count so that I can request a ticket.	5	P0
Sprint 1	Input Validation	As a commuter, I want my inputs validated so that I cannot submit an invalid booking.	3	P0
Sprint 2	Fare Calculation	As a commuter, I want the fare calculated automatically so that I know the exact cost before paying.	8	P0
Sprint 2	QR Ticket Generation	As a commuter, I want a QR-coded digital ticket so that I can use it at the gate without a token.	8	P0
Sprint 3	Notifications	As a commuter, I want instant SMS/Email confirmation so that I have proof of my booking.	5	P0
Sprint 3	Booking History	As a commuter, I want to view past bookings so that I can track my travel and spending.	3	P1
Sprint 4	Payment Gateway	As a commuter, I want to pay via UPI/card/wallet so that I can complete booking instantly.	8	P1
Sprint 4	Cancellation & Refund	As a commuter, I want to cancel a booking and get refunded so that I'm not charged for unused tickets.	5	P1

2.3 Milestones & Timeline



Milestone Schedule

Milestone	Target Timeline	Deliverable
Problem Definition & Empathy Mapping	Week 1	Problem statement, empathy map, prioritized idea list
Requirement Analysis	Week 2	Functional & non-functional requirement specification
Solution Design	Week 3	Data model, workflow design, UI/catalog form design
Project Planning	Week 3	Sprint plan, timeline, resource & risk plan
Sprint 1–4 Development	Week 4–7	Catalog item, fare logic, QR ticketing, notifications
Testing & Deployment	Week 8	Tested, deployed Smart Metro Ticket Booking System

2.4 Team Roles & Resource Allocation

Role	Responsibility
Project Lead	Overall coordination, sprint tracking, and milestone sign-off.
Business Analyst	Requirement gathering, problem statement, and empathy mapping.
ServiceNow Developer	Service Catalog item, Flow Designer logic, and system integrations.
QA / Tester	Test case design and validation of booking, fare, and ticketing logic.
UI/UX Reviewer	Catalog form usability, accessibility, and end-to-end experience checks.

2.5 Risk Management

Risk	Impact	Mitigation
Fare calculation errors	High	Unit-test Flow Designer logic across multiple station pairs before release.
Payment gateway integration delays	Medium	Kept as P1 (non-blocking); MVP functions independently of payment.
Notification delivery failure	Medium	Automatic SMS-to-Email fallback configured.
Scope creep beyond MVP	Medium	Strict adherence to the P0 prioritization matrix for MVP scope.
Timeline slippage	Low	Weekly sprint review with a buffer week before final deployment.

2.6 Planning Summary

The project plan breaks the Smart Metro Ticket Booking System into four focused sprints covering booking, fare calculation, ticketing, notifications, payments, and cancellations. Clear role ownership and an identified risk mitigation plan ensure the MVP can be delivered on the ServiceNow platform within an eight-week timeline, with core P0 capabilities prioritized for early, reliable delivery.